Example 7.66

Find the asymptotes of the function $f(x)=\frac{1}{x}$.

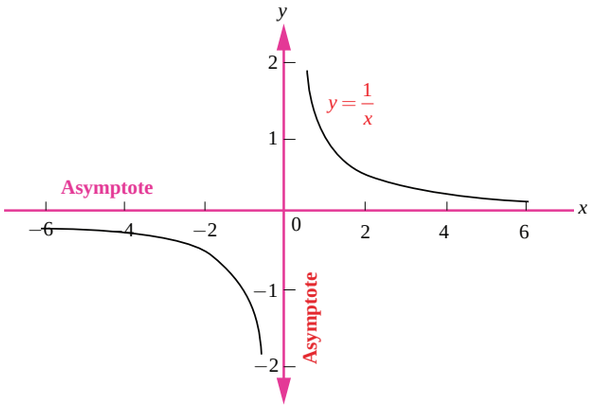


Fig.7.32

Solution

We have, $\lim_{x\to 0^-}\frac{1}{x}=-\infty$ and $\lim_{x\to 0^+}\frac{1}{x}=\infty$. Hence, the required vertical asymptote is $x=0$ or the y-axis.

As the curve is symmetric with respect to both the axes, $y=0$ or the x-axis is also an asymptote. Hence this (rectangular hyperbola) curve has both the vertical and horizontal asymptotes.

Example 7.67

Find the slant (oblique) asymptote for the function $f(x)=\frac{x^2-6x+7}{x+5}$.

Solution

Since the polynomial in the numerator is a higher degree (2nd) than the denominator (1st), we know we have a slant asymptote. To find it, we must divide the numerator by the denominator. We can use long division to do that:

$$
\begin{array}{r}
x-11 \\
x+5\overline{)\,x^2-6x+7} \\
\underline{x^2+5x} \\
-11x+7 \\
\underline{-11x-55} \\
62
\end{array}
$$

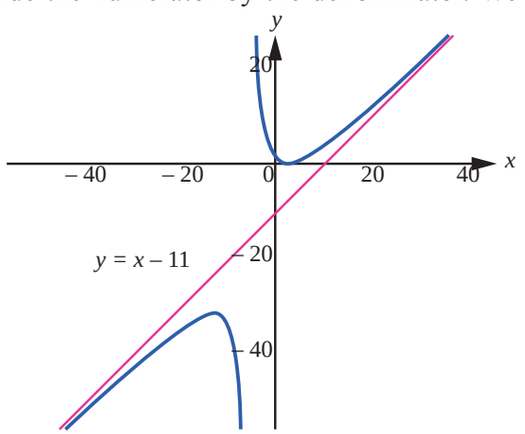


Fig.7.33

Notice that we don't need to finish the long division problem to find the remainder. We only need the terms that will make up the equation of the line. The slant asymptote is $y=x-11$.

As you can see in this graph of the function, the curve approaches the slant asymptote $y=x-11$ but never crosses it:

Example 7.68

Find the asymptotes of the curve $f(x)=\frac{2x^2-8}{x^2-16}$.

Solution

As $\lim_{x\to -4^+}\frac{2x^2-8}{x^2-16}=-\infty$ and $\lim_{x\to 4^+}\frac{2x^2-8}{x^2-16}=\infty$.

Therefore $x=-4$ and $x=4$ are vertical asymptotes.

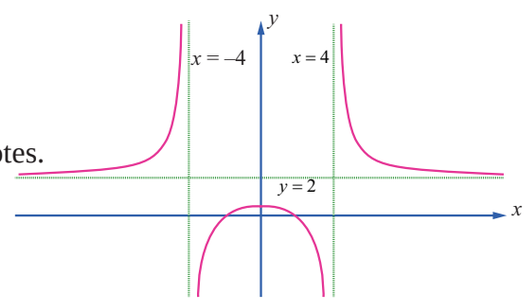


Fig.7.34

$$\text{As } \lim_{x\to\infty}\frac{2x^2-8}{x^2-16}=\lim_{x\to\infty}\frac{2-\frac{8}{x^2}}{1-\frac{16}{x^2}}=2$$